

William Rodriguez

Two-House Buyer Recommendations

Data-Driven Home Recommendations Based on Market Hypotheses

Done by Mohammad Bajbouj & Tim Pfeifer

Contents

1. Workflow Overview
2. H1 Analysis, Recommendations & Charts
3. H2 Analysis, Recommendations & Charts
4. H3 Analysis, Recommendations & Charts
5. Two-House Strategy Summary

Workflow

Data Analysis Pipeline



William's Targets:

- H1: Data-Driven Home Recommendations Based on Market Hypotheses
- H2: Testing whether lot size grows the farther a home is from downtown
- H3: 2+ Floors, Urban

H1: Price vs Renovation Status

Country House Analysis

Testing whether non-renovated rural homes are cheaper, controlling for size

H1 Analysis

Price vs Renovation - Findings

Key Findings

- Rural houses only (7 zip codes, by density)
- Effect direction varies by size
- Medium homes: \$5K pricier (non-reno)
- Large homes: \$5K cheaper (non-reno)
- Small homes: \$3K pricier (non-reno)

→ **Conclusion: data does not support H1 - no tier is significant, and the effect direction flips between tiers**

Key Metrics

Test Used	Mann-Whitney U
Medium n	354 houses
p-value	0.918
Significant?	No
Median Gap	\$5,000
Rural Zips	7 zips (density)

H1 Analysis

House Selection Criteria

Property Targets

- ✓ Location: Rural (quiet, private)
- ✓ Renovation: Non-Renovated
- ✓ Size: Medium (1,570-2,280 sqft)
- ✓ Sorted by price (best value)
- ✓ Matches "space over polish" persona

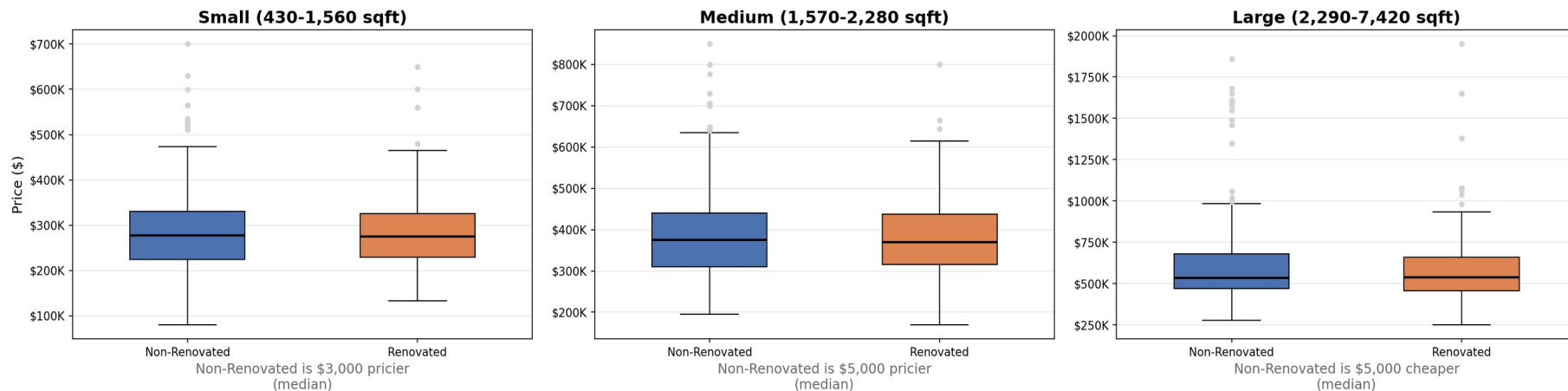
Selection Criteria

Criteria	Target
Location	✓ Rural
Renovation	✓ Non-Renovated
Size Range	✓ Medium
Zipcode	✓ 98022
Homes Found	✓ 2

Price by Renovation Status (by Size)

H1: Non-Renovated Rural Homes Are Not Consistently Cheaper

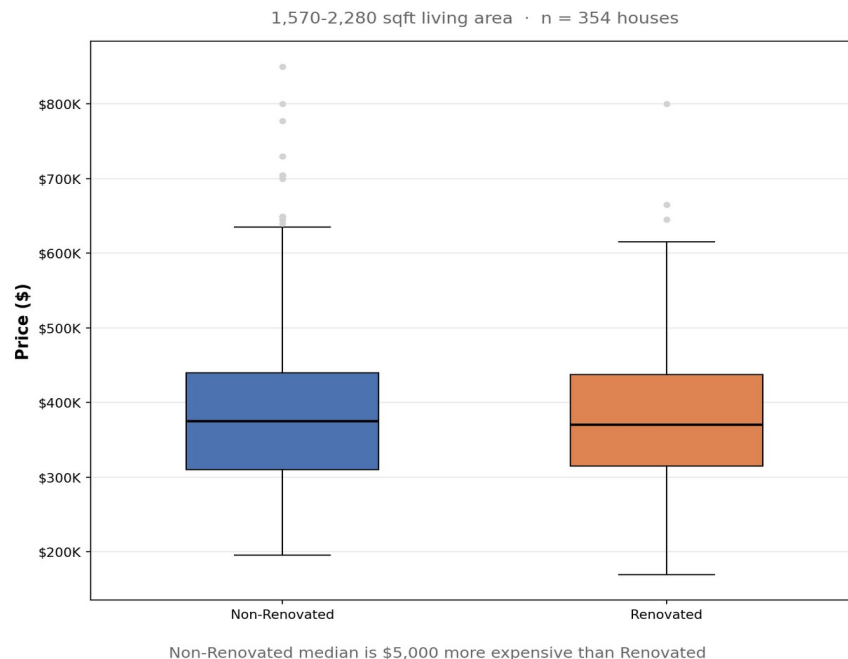
Median price by renovation status, controlling for living-area size (density-based Rural zip codes)



Effect direction is mixed across Small / Medium / Large rural homes

Price by Renovation Status (Medium Only)

H1: Price by Renovation Status - Medium-Size Rural Homes



Focus view: the size band actually used for William's two candidates

Two Rural Homes for William

House #1 - Zip 98022

Price	\$195,000
Living Area	1,580 sqft
Bed / Bath	3 / 1.75
Year Built	1979
Lot Size	7,875 sqft

Why William will love it:

- ✓ Rural, quiet, no close neighbors
- ✓ Non-renovated = lower price
- ✓ Tied for lowest price in candidate pool

House #2 - Zip 98022

Price	\$195,000
Living Area	1,570 sqft
Bed / Bath	3 / 1.75
Year Built	1991
Lot Size	8,459 sqft

Why William will love it:

- ✓ Newer build, slightly larger lot
- ✓ Same price as House #1
- ✓ Tied for lowest price in candidate pool

Housing Density Identifies Rural vs Urban

- Earlier attempts using single features (lot size, distance from downtown, sales per zip) didn't reliably separate Rural from Urban.
- Instead, we use housing density: sales per km², from each zip code's own lat/long bounding box - no external data needed.
- The Rural/Urban split point also comes from the data itself: the largest natural gap in the density distribution, at 1.04 houses/km².
- This identifies 7 Rural zip codes: 98010, 98014, 98019, 98022, 98024, 98045, 98070.
- Checked against the official RUCA classification (for validation only): 80% recall, 94.3% accuracy - beating the 92.9% “always Urban” baseline.
- Conclusion: housing density, derived entirely from the original dataset, is a real way to identify Rural vs Urban zip codes.

H2: Lot Size vs Distance From Downtown Seattle

Testing whether lot size grows the farther a home is from downtown

H2 Analysis

Distance vs Lot Size - Findings

Key Findings

- Positive correlation confirmed
- Effect: +1,440 sqft/mile out
- Highly significant ($p < 0.001$)
- ~20,900 houses analyzed
- H2 hypothesis confirmed

Key Metrics

Distance	Avg Lot Size
5 mi	~5,000 sqft
10 mi	~14,400 sqft
15 mi	~19,000 sqft
20 mi	~23,600 sqft
Correlation	$r = 0.2435$

H2 Analysis

House Selection Criteria

Two Distance Tiers

- Suburban: ~10-15 miles out
- Rural: 15+ miles out
- Pick largest lot in each band
- Both Non-Renovated
- Two distinct price/space tiers

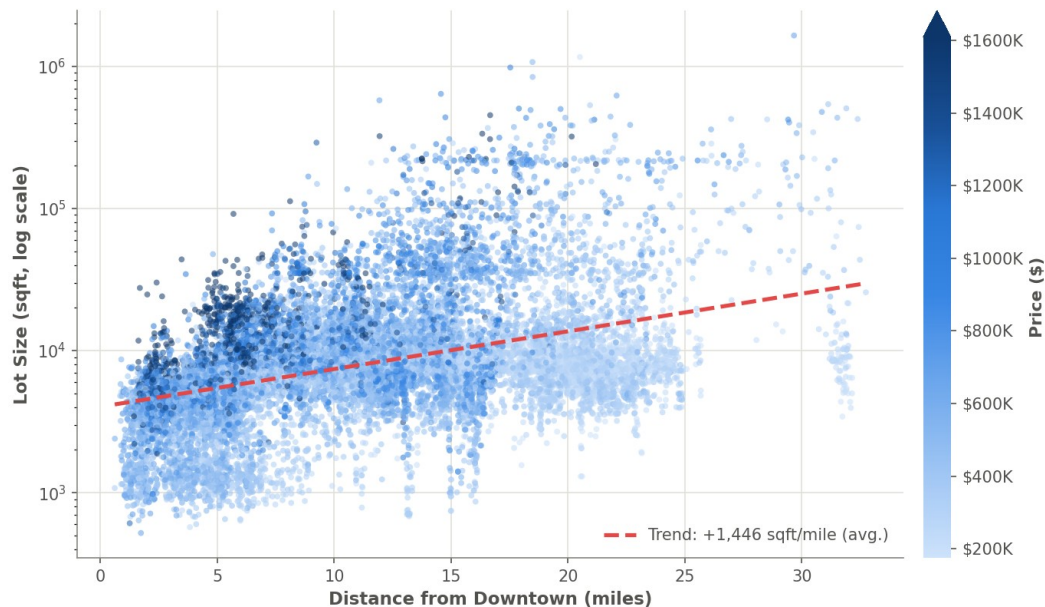
Selection Criteria

Criteria	Target
Suburban Dist.	✓ ~14.6 mi
Rural Dist.	✓ ~29.7 mi
Suburban Lot	✓ 641,203 sqft
Rural Lot	✓ 1,651,359 sqft
Homes Found	✓ 2

Lot Size Grows With Distance (Colored by Price)

H2: Lot Size Grows With Distance From Downtown Seattle

Color = sale price · n = 20,896 houses

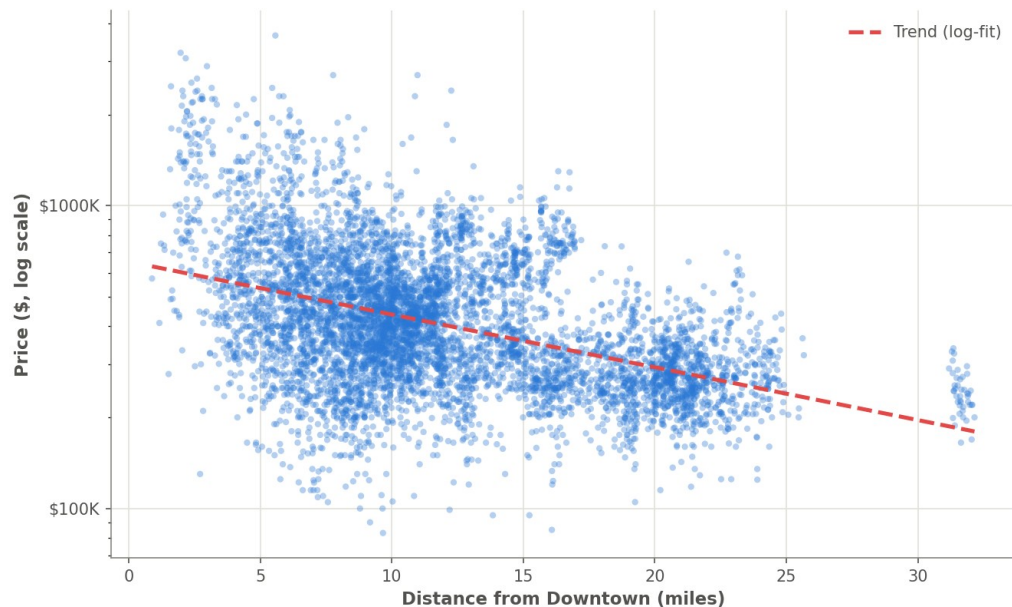


An interactive Seattle map version (zoomable) is also available in [work](#) and [report/figures/](#)

Price vs Distance (Medium Lot-Size Homes)

H2: Price vs Distance - Medium Lot-Size Homes Only

Lot size 6,001-9,360 sqft (middle tertile) · n = 6,759 houses



Among similarly-sized lots, price still drops sharply the farther out you go

Explore the Map Yourself

[H2_plot_lotsize_geographic_map.html](#)

Click to open (works in PowerPoint/Keynote)

Every house in Seattle, zoomable and pannable.

Toggle the color between Lot Size and Price.

Hover any point for price, lot size, and distance.

If the link doesn't open (common in PDF viewers like Preview):

Open Finder → navigate to the project folder → work_and_report → figures
then double-click H2_plot_lotsize_geographic_map.html

Suburban and Rural Options

Suburban - Zip 98070

Price	\$849,900
Distance	14.56 mi
Lot Size	641,203 sqft
Living Area	2,280 sqft
Bed / Bath	2 / 2

Why William will love it:

- ✓ Quiet and peaceful, not too far
- ✓ Large lot for outdoor space
- ✓ Confirmed by H2: closer-in large lot

Rural - Zip 98022

Price	\$700,000
Distance	29.66 mi
Lot Size	1,651,359 sqft
Living Area	1,300 sqft
Bed / Bath	4 / 1

Why William will love it:

- ✓ Maximum quiet and solitude
- ✓ Huge lot: maximum privacy
- ✓ Cheaper than the suburban option

H3: Floors vs Lot Size

Urban City House Analysis

Testing whether multi-floor homes sit on smaller, more compact lots

H3 Analysis

Floors vs Lot Size - Findings & Criteria

Key Findings

- Negative correlation: $r = -0.316$
- Highly significant: $p < 0.0001$
- 18,404 Urban houses analyzed
- 4,706 houses match criteria
- Compact lots = vertical living

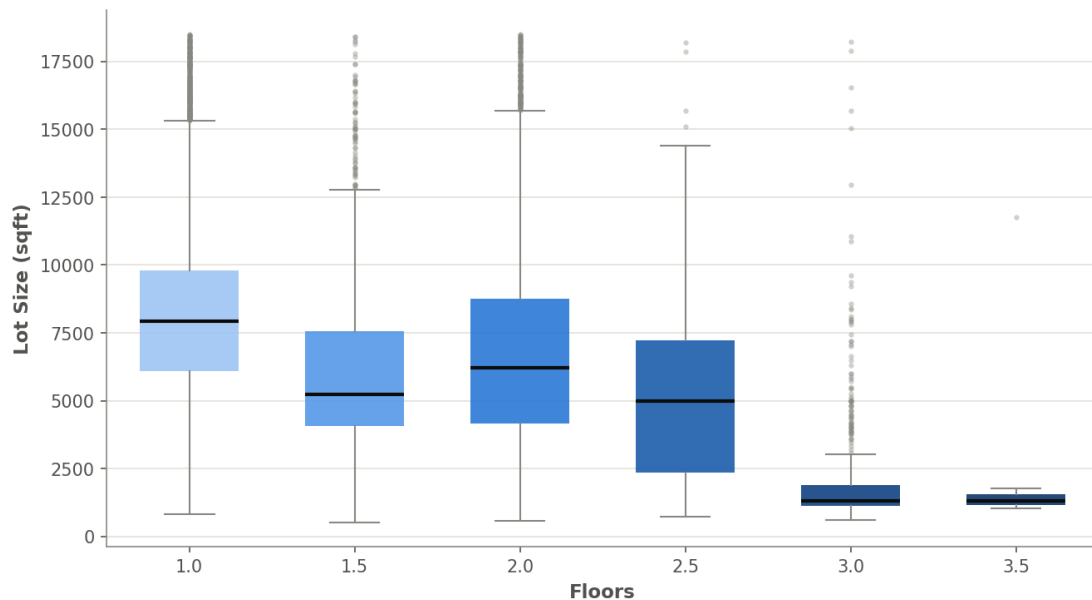
Selection Criteria

Floors	✓ ≥ 2
Lot Size	✓ $\leq 7,200$ sqft (median)
Location	✓ Urban (RUCA 1)
Sort Order	✓ Price descending
Assumption*	✓ Pricier sells faster
Homes Found	✓ 2 (tied at \$2.95M)

Lot Size by Floor Count (Urban Houses)

H3: More Floors Goes With Smaller Lots (Urban Houses)

Spearman $r = -0.316$ · $p < 0.0001$ · $n = 18,404$ houses



More floors goes with smaller lots - $r = -0.316$, $p < 0.0001$, $n = 18,404$

Two Compact City Homes

House #1 - Zip 98107

Price	\$2,950,000
Floors	3.0
Lot Size	5,722 sqft
Living Area	4,340 sqft
Bed / Bath	5 / 4.5

Why William will love it:

- ✓ Genuinely vertical, compact footprint
- ✓ Well below the Urban median lot
- ✓ Tied for highest price - assumed fastest sale

House #2 - Zip 98199

Price	\$2,950,000
Floors	2.0
Lot Size	5,884 sqft
Living Area	4,470 sqft
Bed / Bath	4 / 4.25

Why William will love it:

- ✓ Slightly larger living area
- ✓ Well below the Urban median lot
- ✓ Tied for highest price - assumed fastest sale

Two-House Strategy Summary

William's Country House + City House

Country House - H1

- Rural, Non-Renovated
- Medium size preferred
- \$195K (both candidates)

Country House - H2

- 10mi+ = bigger lots
- Suburban or Rural tiers
- \$700K - \$850K range

City House (H3)

- 2+ floors, compact lot
- Urban core (RUCA 1)
- Candidates: tied at \$2.95M

Q&A

Thank you for listening and paying attention